INCH-POUND

MIL-DTL-22244F (OS)
7 February 2012
SUPERSEDING
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27 May 1997

DETAIL SPECIFICATION

CARTRIDGE, IMPULSE, MARK 23 MOD 0

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the Mark 23 Mod 0 Impulse Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in Sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-D-21625	Design and Evaluation of Cartridge Actuated Devices
MIL-DTL-83420	Wire Rope, Flexible, for Aircraft Control

DEPARTMENT OF DEFENSE STANDARDS

DOD-STD-2101	Classification of Characteristics

Comments, suggestions, or questions on this document should be addressed to Department of the Navy, Indian Head Division, NSWC, Code E12P, Document Control, 4123 Artisans Court, Suite 103, Indian Head, Maryland 20640-5085, OFFICIAL BUSINESS, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/.

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(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia PA 19111- 5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

DL491837	Cartridge, Impulse, Mark 23 Mod 0
1861735	Unloaded Shell Assembly

PUBLICATIONS

Naval Sea Systems Command (Code Ident 53711)

OP 5	Ammunition and Explosives Ashore

(Application for copies should be addressed to the Commanding Officer, Naval Publications and Forms Center, Attn: 1053, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 545	Image Quality in Direct Thermal Neutron Radiographic Examination

(Copies of this document are available online at http://www.astm.org/ or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Conformance to documents. The Mark 23 Mod 0 Impulse Cartridge covered by this specification shall be manufactured in accordance with DL491837.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the

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specifications referred to on the respective drawings. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Bridgewire. The bridgewire shall be soldered in place as specified on Drawing 1861735.

3.5 Bridgewire continuity. The bridgewire shall glow when subjected to the continuity test (see 4.5.4).

3.6 Unloaded shell assembly leakage. The unloaded shell assembly shall not leak (see 4.5.5).

3.7 Propellant charge.

3.7.1 Quantity. All propellant charges shall be determined by weight rather than by volume measurement.

3.7.2 Smokeless powder: handling, and loading. The smokeless powder shall be subjected to no unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.7.3 Smokeless powder, condition. The smokeless powder used shall exhibit no evidence of decomposition (see 4.5.1).

3.8 Cartridge inspection.

3.8.1 Bridge circuit resistance. The resistance of the bridge circuit in each completely assembled cartridge, when measured electrically (see 4.5.3), shall be not less than 3.0 ohms nor greater than 6.5 ohms.

3.8.2 N-ray. Cartridges shall be free of imperfections when N-rayed (see 4.5.6).

3.9 Ballistic requirements. Cartridges shall meet the ballistic requirements specified in 3.9.1 through 3.9.3 when subjected to the test of 4.5.7.

3.9.1 Output. Cartridges shall furnish sufficient power to completely sever the steel cable specified in 4.5.7.

3.9.2 Misfire. There shall be no misfires (see 4.5.7.2).

3.9.3 Hangfire. There shall be no hangfires (see 4.5.7.3).
3.10 Workmanship. Cartridges shall be free of the following defects: burrs, dents, incomplete or damaged threads, improper crimp, perforated closure disc, bent pins, or other defects which may prevent entry of the cartridge into the firing chamber, or affect resistance of the cartridge to moisture. All cartridge marking and identification shall be verified as proper (see 4.5.2).

4. VERIFICATION

4.1 Classification of inspections.

a. First article inspection (see 4.3)

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b. Quality conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5.

4.3 First article inspection. First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2). A first article sample acceptable for environmental and function testing shall be considered a sample which has met the requirements of 3.2 through 3.8.2 and 3.10. First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be from the same sources of supply.

4.3.1 Sample size. The first article sample shall consist of 66 cartridges plus retain samples.

4.3.2 Inspection routine. Sixty-six cartridges shall be expended in the tests listed below and a quantity, specified in the contract, of retain samples used for investigative purposes. Testing shall consist of the following tests as prescribed in MIL-D-21625F: 40-foot drop (6); temperature and humidity cycling (12), except that 14 days shall be used in lieu of 28 days; vibration (12); -65°F temperature (12); 160°F temperature (12); 70°F temperature (12). The number in parenthesis denotes the number of cartridges allocated for each test. No cartridge shall produce results which fail to meet the requirements of 3.9.1 through 3.9.3. Firings shall be conducted in the test fixture of 4.5.7.

4.3.3 Failure. Failure of any cartridge shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the tests and examinations specified in table I and verification of the characteristics classified on the drawings of DL491837.

4.4.1 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not less than 281 and not greater than 10,000. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component lot may be used in more than one cartridge production lot.

4.4.2 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with table II.

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TABLE I. Quality conformance tests and examinations.

Item	Nature of test	Requirement	Test method	Classification (DOD-STD-2101)
Bridgewire	Soldering	3.4	—	(M101)
Bridgewire	Continuity	3.5	4.5.4	(C1)
Unloaded shell assembly	Leakage	3.6	4.5.5	(C2)
Propellant charge	Type and weight	3.7.1	—	(C3)
Smokeless powder	Condition	3.7.3	4.5.1	(C4)
Cartridge	Visual inspection	3.10	4.5.2	(M102)
Cartridge	Bridgewire resistance	3.8.1	4.5.3	(C5)
Cartridge	N-ray	3.8.2	4.5.6	(C6)
Cartridge	Ballistics	3.9	4.5.7	(C7)
Cartridge	Preservation-packaging, packing, and marking	Section 5, Contract	4.5.2	(M103)

TABLE II. Quality conformance sample size.

Lot size	Sample size	Retained sample
51 to 90	13	*
91 to 150	20	*
151 to 280	32	*
281 to 500	50	*
501 to 1200	80	*
1201 to 3200	125	*
3201 to 10,000	200	*
10,001 to 35,000	315	*

* Quantity to be specified in the contract.

4.4.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

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4.5 Tests.

4.5.1 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

4.5.2 Visual examination. All cartridges shall be visually examined. Cartridges having any of the defects of 3.10 shall be cause for rejection. The preservation-packaging, packing, and marking of the cartridges shall be examined for conformance to section 5 and the contract (see 6.3).

4.5.3 Bridge circuit resistance. Conformance with the bridge circuit resistance of 3.8.1 shall be ascertained on all completely assembled cartridges by means of a test circuit which limits the bridge circuit current to 50 milliamperes, maximum. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument which is known to be accurate to within 0.005 ohm in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirement of 3.8.1 shall be cause for rejection. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and connector.

4.5.4 Bridgewire circuit continuity. After soldering the bridgewire, the unloaded shell assembly (Drawing 1861735) shall be thoroughly cleaned to remove flux and contamination. After cleaning, the bridgewire shall be flashed with an electric current using a voltage source of 6 volts ac or dc to produce a visible red glow of the bridge and ensure continuity. A suitable resistor may be used to regulate the current to the bridge. Flashing is a necessary manufacturing operation and shall be conducted on each loaded assembly. Assemblies with bridgewires which fail to glow shall be cause for rejection. Any faulty assemblies may be reworked by replacement of the bridges.

4.5.5 Unloaded shell assembly leakage. Apply not less than 10 psi of air pressure internally to all unloaded shell assemblies (Drawing 1861735) which have passed the continuity test of 4.5.4 and check for leaks. All assemblies that show evidence of leakage shall be cause for rejection.

4.5.6 Radiographic inspection. All cartridges shall be radiographically examined in accordance with ASTM E 545. Cartridges having any observable imperfections in assembly shall be cause for rejection. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive numerical order beginning with the number 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the N-ray. Defective cartridges found by the vendor shall be clearly identified on the N-ray and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridge with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings.

4.5.6.1 Sample radiographic plate. A sample radiographic plate and technique shall be submitted for n-ray by the contractor for approval prior to submitting radiographs for the production lot. The production lot radiographs shall be made using the identical technique utilized to produce the approved sample plate.

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4.5.6.2 N-ray examination. Each cartridge shall be n-rayed in accordance with ASTM E 545, to confirm the presence of the propellant, non-metal parts, no foreign hydrogenous material and proper assemby of pyrotechnic/non-metallic materials. Any observable imperfections shall be cause for cartidge rejection

4.5.7 Ballistic test. The number of cartridges selected from each production lot for the ballistic test shall be in accordance with table II. The cartridges shall be conditioned for not less than 6 hours at 70 ± 5°F and fired in a cable cutter, Mine Safety Appliance Company Part No. TA 97116 or Part No. TA 97999, or equal. A 3/16-inch-diameter steel cable in accordance with MIL-DTL-83420, Type I, Composition B, shall be properly positioned in the cable cutter for cutting. The cable cutter shall be adapted to measure the time from actuation of the firing mechanism to complete severance of the cable. The firing current shall be 4.0 + 0.5, -0.0 amperes. In addition to the sample of cartridges for the above test, additional cartridges shall be furnished from each production lot for investigative purposes. The quantity of this investigative sample shall be as specified in the contract.

4.5.7.1 Criteria of acceptability. The failure of any cartridge to completely sever the steel cable shall be cause for rejection of the lot represented.

4.5.7.2 Misfire. The failure of any cartridge to fire shall be cause for rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.5.7.3 Hangfire. One or more hangfires (see 6.4.5) shall be cause for rejection of the lot represented.

4.5.7.4 Retest. There shall be no retests.

4.5.7.5 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridge, original test results should be discarded and that portion of the test reconducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements will be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The Mark 23 Mod 0 Impulse Cartridge is used to actuate a helicopter-rescue hoist cable cutter (MSA Part No. TA 97116 or 97999, or equal) to jettison a load in an emergency. Since the cartridge was developed for use in military helicopters, there is no commercial application.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, date and revision of the specification and of the top assembly drawing.

b. If required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2).

c. Date and place of delivery.

d. Inspection conditions, if other than as specified (see 4.2).

e. Lot size, if other than as specified (see 4.4.1).

f. Sampling, if other than as specified (see 4.4.2).

g. Packaging requirements and any special marking (see 5.1, 6.3, and contract).

h. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" OP 5 are applicable.

6.3 Definitions.

6.3.1 Hangfire. A hangfire is defined as a firing in which the elapsed time between actuation of the firing mechanism and complete severance of the steel cable exceeds 0.250 second.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

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6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:
Navy – OS

Preparing activity:
Navy - OS
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